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British Standard

# Wood poles for overhead power and telecommunication lines

## Part 1. Specification for softwood poles

Poteaux en bois pour lignes électriques et télégraphiques aériennes — Spécifications  
Partie 1. Poteaux en bois tendre

Holzmaße für Hochspannungs- und Telefonleitungen  
Teil 1. Maße aus Weichholz



British Standards Institution

## Foreword

This British Standard, having been prepared under the direction of the Timber Standards Committee, supersedes BS 1990 : 1971 which is withdrawn. The revision is being undertaken in two Parts; Part 1 covering softwood poles and Part 2\* which will cover hardwoods. The first standard for wood poles for overhead lines was published in 1953. A metric version of the 1953 edition was prepared and published in 1971 superseding the imperial version in 1974.

The difference between this edition and the 1971 edition is that this standard extends the range of timber species for which data are available, specifies water-borne preservative treatments in addition to pressure creosoting and gives a typical layout of markings. This standard is based primarily upon the predominant species in use, namely redwood or Scots pine (*Pinus sylvestris*) from Northern Europe including the United Kingdom.

The growth characteristics of timbers and their freedom from decay and other defects are of importance in the determination of the mechanical strength of poles for assessing the safe loads in service. The calculated average breaking loads of redwood or Scots pine poles both unstayed and stayed are included in appendix A.

Appendix B describes a method of determining the moisture content of poles and appendix C describes a method for determining the extent of preservative penetration.

Appendix D describes a method for establishing the strength characteristics for the species listed in this standard.

The attention of pole users is drawn to the difficulty of defining and working to precise requirements on all aspects of a natural product such as timber. Some properties have of necessity been specified in general terms only. It is therefore recommended that the acceptance of poles, and the supervision of seasoning tests and of preservation processes should be assigned only to qualified and experienced inspectors.

**Compliance with a British Standard does not of itself confer immunity from legal obligations.**

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\*In course of preparation.

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# Specification

## 1 Scope

This British Standard specifies requirements for solid softwood poles for power and telecommunication lines (see clause 3).

NOTE. The titles of the publications referred to in this standard are listed on the inside back cover.

## 2 Definitions

For the purposes of this British Standard the definitions given in BS 565, BS 4261 and BS 6100 : Section 4.1 apply.

## 3 Timbers

Poles shall be made from the following species.

- |                            |                                  |
|----------------------------|----------------------------------|
| (a) Redwood or Scots pine* | ( <i>Pinus sylvestris</i> )      |
| (b) Corsican pine          | ( <i>Pinus nigra</i> )           |
| Douglas fir                | ( <i>Pseudotsuga menziesii</i> ) |
| Dunkeld (hybrid) larch     | ( <i>Larix eurolepis</i> )       |
| European larch             | ( <i>Larix decidua</i> )         |
| European (Norway) spruce   | ( <i>Picea abies</i> )           |
| Japanese larch             | ( <i>Larix kaempferi</i> )       |
| Lodgepole pine             | ( <i>Pinus contorta</i> )        |
| Sitka spruce               | ( <i>Picea sitchensis</i> )      |
| Western red cedar          | ( <i>Thuja plicata</i> )         |

NOTE. The preferred dressed sizes of poles should be as given in table 1.

Poles of redwood should support the loads given in table 2 and table 3 derived from the calculations in accordance with appendix A. Poles of the species listed under item (b) should support the loads given in table 2 and table 3 modified by the appropriate factors given in table 4.

## 4 Felling

Trees other than spruce, grown in Northern Europe, shall be felled between 1 November and the following 31 March. They shall be sawn off as close to the ground as possible and no timber shall be removed, trimmed or cut from the butt end so as to reduce its natural size. The ends shall be sawn to give a flat section and branches shall be dressed down flush with the trunk. Bark shall be removed as soon as practicable after felling and the poles shall then either be stacked in open formation clear of the ground, or stored in water that is free from marine borer activity.

## 5 Sizes and permitted tolerances

NOTE. This standard is applicable to all sizes of pole, however, it will generally be more economic to specify the preferred dressed sizes given in table 1.

### 5.1 Measurement

Many nominally round poles have a natural elliptical cross section; the cross section measurements of such poles shall be of the minor axes.

### 5.2 Tolerance on length

The tolerance on length shall be  $\pm 1\%$  of the specified length.

## 6 Limiting characteristics

NOTE. Some limiting characteristics may be eliminated by cross cutting and/or dressing to produce an acceptable pole.

### 6.1 Straightness

Sweep or crook shall be permitted to the extent that a straight line from the centre of the top to the centre of the pole 1.5 m from the butt remains within the pole.

### 6.2 Knots

At any cross section along the length of a pole no knot with a diameter, measured at right angles to the length of the pole, greater than one-fifth of the circumference shall be permitted, and the sum of knot diameters at the cross section shall not exceed one-quarter of the circumference†.

### 6.3 Fissures

Cracks across poles shall not be permitted.

NOTE. Seasoning checks and shakes along the grain are expected and are not recognized as defects.

### 6.4 Ring and star shakes

The tip shall be free from ring or star shakes but one complete ring or one star shake is acceptable at the butt provided that no part extends to within 50 mm from the periphery.

### 6.5 Slope of grain

Spiral grain shall not exceed a slope of 1 in 6.

\* Hereinafter referred to as redwood.

† The limitations on knots are based on past practice and satisfactory performance. Suitable test results relating knot size to performance are not available.



### 6.6 Inbarks

Inbarks shall be permitted with limitations on their length, width and depth. Maxima are given for each dimension, but when assessing inbarks, the maxima shall only be permitted for one measurement, with half the value given for the other two.

The maximum dimensions of inbarks shall be as follows.

Length: 4 x diameter of pole

Width: 0.5 x diameter of pole

Depth: 12 mm

### 6.7 Insect holes

Pinholes and boreholes shall be permitted, provided there is no active infestation and the strength of the pole is not affected by the degree of damage.

### 6.8 Decay

Rot or punks shall not be permitted.

### 6.9 Abnormal defects

Other defects, such as mechanical damage, shall be permitted provided that their effect on the structural strength of the pole is likely to be no greater than the effect of any other limiting characteristic.

### 6.10 Sapstain

Blue stain in the sapwood of softwoods shall not be considered a defect.

## 7 Stacking and preparation of poles at depots

All poles other than spruce shall be stacked in open formation, before and after dressing, on suitable skids so that the lowest timber of each stack is at least 250 mm above the ground. Sufficient dunnage or cross timbers shall be used to distribute the total mass without imposing undue strain on the poles at the lower layers. The surface of the seasoning ground under and around the stacks shall be well drained and kept free from bark, shavings, grass and weeds. All poles shall be machine dressed to remove the cambium layer before preservation. All drilling, notching and cutting of poles shall be completed before preservation.

## 8 Moisture content

Preservation of species other than spruce shall not take place until the average moisture content of each batch of poles is reduced to not more than 28 %. No individual pole in the batch shall have a moisture content greater than 30 %.

The moisture content of the poles shall be determined prior to treatment by the oven test method described in appendix B or by the use of moisture meters. When using the latter, any measurements over 25 % shall be confirmed by the oven test method.

Poles for preservation shall be protected against heavy and continuous rain.

NOTE. Seasoning is deemed to commence at the time of felling, and under average weather conditions light, medium and stout poles should be seasoned for at least 6, 9 and 12 months respectively, before preservation.

## 9 Preservation

NOTE. Long life is a prerequisite for poles and preservative treatment is essential. It is recommended that poles complying with this standard have an average service life of 40 years.

### 9.1 General

The following species shall be pressure impregnated with creosote in accordance with BS 913 or with a copper/chrome/arsenic mixture in accordance with BS 4072.

Redwood  
Corsican pine  
Douglas fir  
Dunkeld larch  
European larch  
Japanese larch  
Lodgepole pine  
Western red cedar

When pressure impregnating in accordance with BS 913, the coal tar creosote used shall comply with type 2 of BS 144 and the average net retention of preservative in a charge shall be not less than 115 kg/m<sup>3</sup>.

When pressure impregnating in accordance with BS 4072, the copper/chromate/arsenic (CCA) mixture shall have a solution strength not less than 3 % m/V.

NOTE 1. When creosoting is carried out using the Rueping (empty cell) process the gross absorption, to obtain full sapwood penetration, is likely to be double the net retention. However, as the depth of sapwood varies from pole to pole and between species, this level of gross absorption is included for guidance only.

NOTE 2. The average net retention figures given in BS 913 and BS 4072 are average values for a whole charge and do not apply to individual poles. Owing to natural variations in the permeability of wood it has to be accepted that the net retention of individual poles may cover a considerable range on either side of any average value specified for the charge. Initial, gross and net absorptions may also vary from that specified due to the variation in the depth of sapwood present and the species being treated.

As the following species cannot be treated with preservative in the normal way by the procedures described in BS 913 and BS 4072, they shall be preserved using a suitable modification of one of the procedures described in those standards, to meet the requirements for full sapwood penetration and net retention.

European (Norway) spruce  
Sitka spruce

### 9.2 Preparation of poles for treatment

Preservative treatment shall not take place until all dressing, notching, pre-cutting and boring has been completed.

NOTE. Fabricated poles ready for treatment are to be protected against heavy and continuous rain (see clause 8).

### 9.3 Extent of penetration of preservative

When examined in accordance with appendix C, all sapwood shall be completely penetrated with preservative.

## 10 Marking

### 10.1 General

Each pole shall have a gouge or brand mark at least 50 mm long, 5 mm wide and 3 mm deep placed 3 m from the butt and shall be marked with the information specified in 10.2 and 10.3 in characters not less than 25 mm high and gouged or branded 5 mm wide and 3 mm in depth. Spacings between codes shall be between 20 mm and 30 mm (see figure 1).

### 10.2 Above the gouge mark

Each pole shall be marked with the following information above the gouge mark:

- (a) length of pole (in m);
- (b) class of pole, i.e. L (light), M (medium) or S (stout) or, alternatively, the diameter (in mm) at 1.5 m from the butt.

NOTE. The ownership mark may also be marked if required.

### 10.3 Below the gouge mark

Each pole shall be marked with the following information below the gouge mark:

(a) species code, i.e.:

|    |                       |
|----|-----------------------|
| CF | Corsican pine         |
| DF | Douglas fir           |
| LP | Lodgepole pine        |
| LX | Larch, Dunkeld        |
|    | Larch, European       |
|    | Larch, Japanese       |
| NS | Norway spruce         |
| SP | Scots pine or Redwood |
| SS | Sitka spruce          |
| WC | Western red cedar     |

(b) country of origin\*, i.e.:

|    |                |
|----|----------------|
| GB | United Kingdom |
| CA | Canada         |
| FI | Finland        |
| NO | Norway         |
| SE | Sweden         |

(c) last two digits of year of preservation.

NOTE. The depot supplying the pole and the name or mark of the preserver may also be marked.

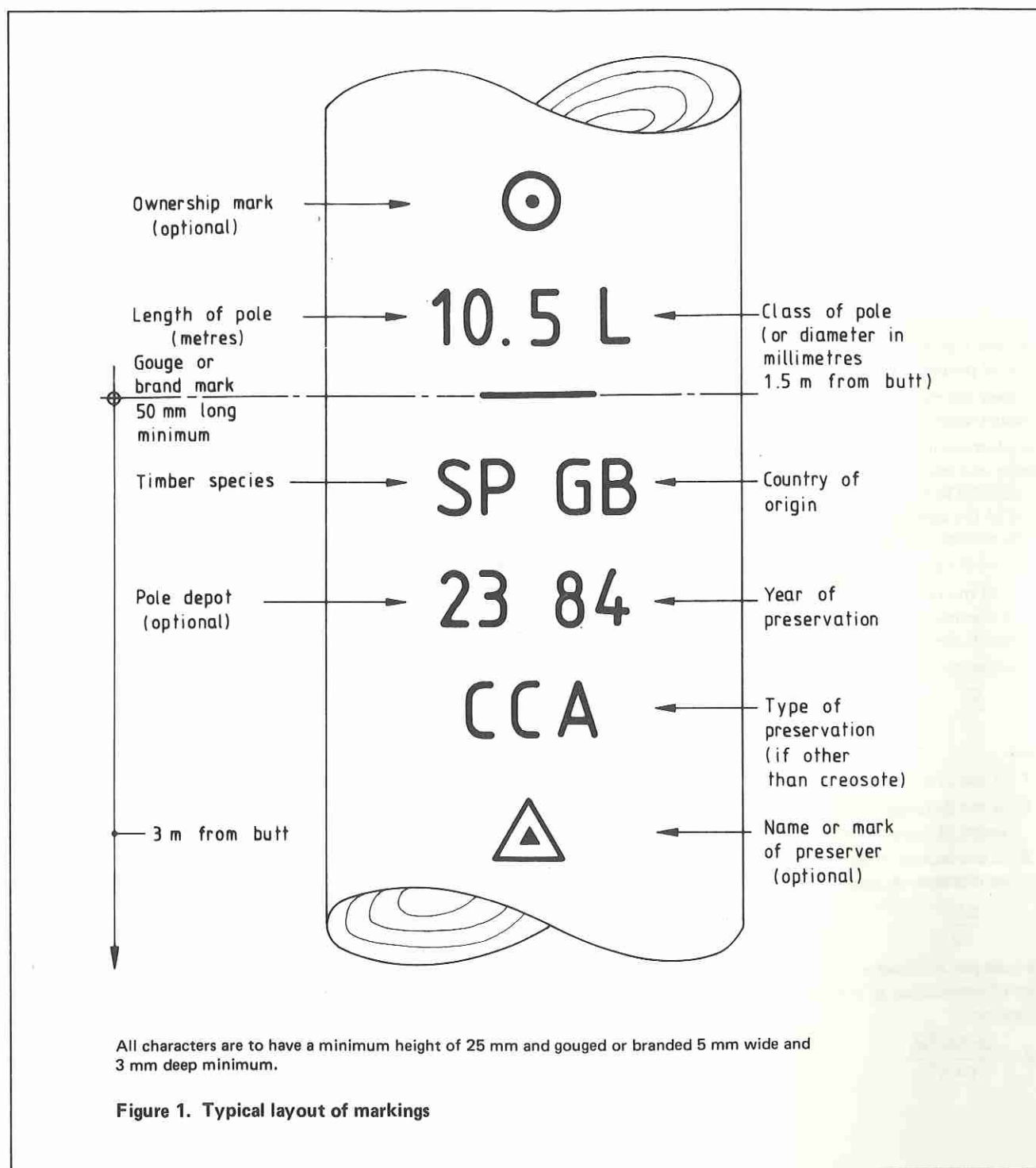
Poles preserved with creosote shall carry no other distinguishing mark. Other forms of preservation shall be denoted by appropriate letters.

### 10.4 The butt end

Each pole shall have the following information impressed on the butt end grain:

- (a) length of pole (in m);
- (b) class of pole, i.e. L (light), M (medium) or S (stout) or, alternatively, the diameter (in mm) at 1.5 m from the butt.

\* For abbreviations of other countries, see BS 5374.





## Appendix A. Design data for the strength of poles, unstayed and stayed

### A.1 Loads

The ultimate load for species other than redwood (i.e. excluding safety factors) for a pole size given in table 1 may be calculated as the product of the appropriate pole load from tables 2 and 3 and the modification factor for the species given in table 4. For other sizes of poles the loads should be calculated individually.

### A.2 Unstayed poles

The loads given in table 2 are assumed to be applied 0.6 m\* from the top of the pole and are based on the following material property values for redwood:

mean ultimate bending stress 53.8 N/mm<sup>2</sup>  
mean modulus of elasticity 10 480 N/mm<sup>2</sup>

The ultimate loads  $F$  calculated according to simple bending theory and assuming a rigidly supported cantilever, correspond to the mean ultimate strength of unsupported poles of the specified sizes. They are based on the strength at the critical cross section which occurs:

- (a) at the groundline; or
- (b) at the point where the diameter is equal to 1.5 times the diameter at the point of application of load, if this point is above the ground line.

The ultimate load  $F$  (in N) is given by the expression:

$$F = \frac{fZ}{l_c}$$

where

- $f$  is the ultimate bending stress (in N/mm<sup>2</sup>)
- $l_c$  is the distance between the critical section and the point of application of the load (in mm);
- $Z$  is the section modulus (in mm<sup>3</sup>) at the critical section of diameter  $d_c$  and is given by

$$\frac{\pi d_c^3}{32}$$

The load per millimetre of deflection,  $P$  (in N/mm), at the point of application of the load is given by the expression:

$$P = \frac{3\pi E d_1^3 d_2}{64 h^3}$$

where

- $E$  is the modulus of elasticity (in N/mm<sup>2</sup>);
- $d_1$  is the diameter at the groundline (in mm);
- $d_2$  is the diameter at the point of application of the load (in mm);
- $h$  is the distance from the groundline to the point of application of the load (in mm).

### A.3 Stayed poles (struts or columns)

Where stays are used, the ability of the pole to resist the crippling loads due to the vertical component of the forces in the stays should be considered. The ultimate crippling loads given in table 3 are for guidance only. They are calculated from the Euler formula for columns, modified to allow for:

- (a) the tapered section of round poles;
- (b) the imperfect rigidity of the ground in which the poles are planted;
- (c) a degree of stability at the top of the pole due to stays and line wires.

The crippling loads  $F_c$  (in N) are given by the expression:

$$F_c = 0.00411 \frac{d_e^4}{l^2}$$

where

- $l$  is the effective length, which is taken as the length between a point 0.3 m below the top and 0.3 m below the groundline and is therefore equal to  $L - D$  (in m).

$d_e$  is the effective diameter (in mm), given by the expression:

$$d_e = d_t + \frac{(L - D + 0.9)(d_b - d_t)}{3(L - 1.5)}$$

where

- $L$  is the full length (in m)
- $D$  is the depth of planting (in m);
- $d_t$  is the top diameter (in mm);
- $d_b$  is the diameter at 1.5 m from the butt (in mm).

The values calculated using these formulae and included in table 3 should be used with caution, especially when the pole is to be set in particularly unstable ground.

NOTE. It is good practice to select the straightest and stoutest poles for stayed poles.

\* If the load is to be applied at a position less than 0.6 m from the top of the pole, tables 2 and 3 are no longer valid, and loads should be calculated for the particular loading condition.



Table 1. Preferred sizes of poles

| Nominal length | Light poles     |      |                                         | Medium poles    |      |                                         | Stout poles     |      | Minimum diameter at 1.5 m from butt end |
|----------------|-----------------|------|-----------------------------------------|-----------------|------|-----------------------------------------|-----------------|------|-----------------------------------------|
|                | Diameter at top |      | Minimum diameter at 1.5 m from butt end | Diameter at top |      | Minimum diameter at 1.5 m from butt end | Diameter at top |      |                                         |
|                | min.            | max. |                                         | min.            | max. |                                         | min.            | max. |                                         |
| m              | mm              | mm   | mm                                      | mm              | mm   | mm                                      | mm              | mm   |                                         |
| 6              | 125             | 150  | 150                                     | —               | —    | —                                       | —               | —    |                                         |
| 7              | 125             | 150  | 160                                     | 140             | 170  | 200                                     | —               | —    |                                         |
| 8              | 125             | 150  | 170                                     | 145             | 180  | 210                                     | —               | —    |                                         |
| 8.5            | 125             | 150  | 180                                     | 150             | 180  | 215                                     | 190             | 240  |                                         |
| 9              | 125             | 150  | 180                                     | 150             | 180  | 220                                     | 190             | 240  |                                         |
| 9.5            | 125             | 160  | 185                                     | 150             | 180  | 225                                     | 190             | 245  |                                         |
| 10             | 125             | 160  | 185                                     | 150             | 185  | 230                                     | 190             | 245  |                                         |
| 10.5           | 125             | 160  | 190                                     | 150             | 190  | 235                                     | 190             | 250  |                                         |
| 11             | 125             | 160  | 195                                     | 150             | 190  | 240                                     | 190             | 250  |                                         |
| 11.5           | 125             | 165  | 200                                     | 150             | 190  | 245                                     | 190             | 250  |                                         |
| 12             | 125             | 165  | 200                                     | 150             | 190  | 250                                     | 190             | 250  |                                         |
| 13             | 130             | 170  | 210                                     | 160             | 200  | 260                                     | 195             | 255  |                                         |
| 14             | —               | —    | —                                       | 160             | 205  | 275                                     | 195             | 255  |                                         |
| 15             | —               | —    | —                                       | 165             | 210  | 290                                     | 195             | 260  |                                         |
| 16             | —               | —    | —                                       | 170             | 215  | 305                                     | 200             | 265  |                                         |
| 17             | —               | —    | —                                       | 180             | 220  | 320                                     | 200             | 265  |                                         |
| 18             | —               | —    | —                                       | 180             | 220  | 330                                     | 200             | 265  |                                         |
| 20             | —               | —    | —                                       | 180             | 230  | 360                                     | 200             | 265  |                                         |
| 22             | —               | —    | —                                       | 190             | 240  | 380                                     | 200             | 265  |                                         |
| 24             | —               | —    | —                                       | —               | —    | —                                       | 200             | 265  |                                         |

NOTE. Maximum top diameters are given only to indicate the range of taper to be expected.

Table 2. Ultimate loads for unstayed poles of redwood

| Nominal length of pole | Assumed depth of planting* | Light poles                             |                                         |                                                                   | Medium poles                            |                                         |                                                                   | Stout poles                             |                                         |                                                                   |
|------------------------|----------------------------|-----------------------------------------|-----------------------------------------|-------------------------------------------------------------------|-----------------------------------------|-----------------------------------------|-------------------------------------------------------------------|-----------------------------------------|-----------------------------------------|-------------------------------------------------------------------|
|                        |                            | Minimum diameter at 1.5 m from butt end | Ultimate load at 0.6 m from top of pole | Load per millimetre of deflection at point of application of load | Minimum diameter at 1.5 m from butt end | Ultimate load at 0.6 m from top of pole | Load per millimetre of deflection at point of application of load | Minimum diameter at 1.5 m from butt end | Ultimate load at 0.6 m from top of pole | Load per millimetre of deflection at point of application of load |
| m                      | m                          | mm                                      | kN                                      | N                                                                 | mm                                      | kN                                      | N                                                                 | mm                                      | kN                                      | N                                                                 |
| 6                      | 1.2                        | 150                                     | 4.39                                    | 9.3                                                               | —                                       | —                                       | —                                                                 | —                                       | —                                       | —                                                                 |
| 7                      | 1.2                        | 160                                     | 4.31                                    | 6.0                                                               | 200                                     | 8.53                                    | 13.5                                                              | —                                       | —                                       | —                                                                 |
| 8                      | 1.5                        | 170                                     | 4.40                                    | 4.8                                                               | 210                                     | 8.29                                    | 10.5                                                              | —                                       | —                                       | —                                                                 |
| 8.5                    | 1.5                        | 180                                     | 4.81                                    | 4.5                                                               | 215                                     | 8.20                                    | 9.1                                                               | 265                                     | 15.35                                   | 21.5                                                              |
| 9                      | 1.5                        | 180                                     | 4.46                                    | 3.6                                                               | 220                                     | 8.15                                    | 7.8                                                               | 275                                     | 15.91                                   | 19.2                                                              |
| 9.5                    | 1.5                        | 185                                     | 4.52                                    | 3.1                                                               | 225                                     | 8.13                                    | 6.8                                                               | 280                                     | 15.66                                   | 16.4                                                              |
| 10                     | 1.5                        | 185                                     | 4.23                                    | 2.6                                                               | 230                                     | 8.13                                    | 5.9                                                               | 285                                     | 15.47                                   | 14.3                                                              |
| 10.5                   | 1.5                        | 190                                     | 4.31                                    | 2.3                                                               | 235                                     | 8.16                                    | 5.3                                                               | 290                                     | 15.33                                   | 12.5                                                              |
| 11                     | 1.8                        | 195                                     | 4.40                                    | 2.3                                                               | 240                                     | 8.18                                    | 5.0                                                               | 295                                     | 15.24                                   | 11.8                                                              |
| 11.5                   | 1.8                        | 200                                     | 4.48                                    | 2.1                                                               | 245                                     | 8.21                                    | 4.5                                                               | 300                                     | 15.15                                   | 10.5                                                              |
| 12                     | 1.8                        | 200                                     | 4.26                                    | 1.7                                                               | 250                                     | 8.23                                    | 4.1                                                               | 305                                     | 15.08                                   | 9.4                                                               |
| 13                     | 1.8                        | 210                                     | 4.46                                    | 1.6                                                               | 260                                     | 8.46                                    | 3.7                                                               | 320                                     | 15.73                                   | 8.3                                                               |
| 14                     | 1.8                        | —                                       | —                                       | —                                                                 | 275                                     | 8.98                                    | 3.3                                                               | 335                                     | 16.24                                   | 7.3                                                               |
| 15                     | 2.1                        | —                                       | —                                       | —                                                                 | 290                                     | 9.60                                    | 3.3                                                               | 350                                     | 16.68                                   | 6.8                                                               |
| 16                     | 2.1                        | —                                       | —                                       | —                                                                 | 305                                     | 10.23                                   | 3.1                                                               | 385                                     | 17.35                                   | 6.2                                                               |
| 17                     | 2.1                        | —                                       | —                                       | —                                                                 | 320                                     | 11.07                                   | 3.0                                                               | 375                                     | 17.20                                   | 5.4                                                               |
| 18                     | 2.1                        | —                                       | —                                       | —                                                                 | 330                                     | 11.14                                   | 2.7                                                               | 390                                     | 17.57                                   | 5.0                                                               |
| 20                     | 2.1                        | —                                       | —                                       | —                                                                 | 360                                     | 11.98                                   | 2.5                                                               | 415                                     | 17.74                                   | 4.2                                                               |
| 22                     | 2.4                        | —                                       | —                                       | —                                                                 | 380                                     | 12.63                                   | 2.3                                                               | 435                                     | 17.49                                   | 3.6                                                               |
| 24                     | 2.4                        | —                                       | —                                       | —                                                                 | —                                       | —                                       | —                                                                 | 470                                     | 18.36                                   | 3.3                                                               |

\* The assumed depth of planting is not based on the necessary foundation design requirements.

Table 3. Ultimate loads for stayed poles (struts) of redwood

| Nominal length | Light poles        |                | Medium poles       |                | Stout poles        |                |
|----------------|--------------------|----------------|--------------------|----------------|--------------------|----------------|
|                | Effective diameter | Crippling load | Effective diameter | Crippling load | Effective diameter | Crippling load |
| m              | mm                 | kN             | mm                 | kN             | mm                 | kN             |
| 6              | 135.6              | 60.3           | —                  | —              | —                  | —              |
| 7              | 139.2              | 45.9           | 164.0              | 89.2           | —                  | —              |
| 8              | 142.1              | 39.7           | 170.0              | 80.7           | —                  | —              |
| 8.5            | 145.7              | 37.8           | 174.5              | 77.8           | 218.3              | 190.4          |
| 9              | 145.6              | 32.8           | 176.2              | 70.4           | 221.8              | 176.8          |
| 9.5            | 147.3              | 30.2           | 177.9              | 64.3           | 223.4              | 160.0          |
| 10             | 147.1              | 26.6           | 179.5              | 59.1           | 225.1              | 146.1          |
| 10.5           | 148.9              | 24.9           | 181.2              | 54.7           | 226.7              | 134.0          |
| 11             | 149.8              | 24.5           | 181.9              | 53.2           | 227.3              | 129.5          |
| 11.5           | 151.5              | 23.0           | 183.6              | 49.6           | 228.9              | 120.0          |
| 12             | 151.5              | 20.8           | 185.3              | 46.6           | 230.6              | 111.7          |
| 13             | 158.1              | 20.5           | 195.1              | 47.5           | 238.9              | 106.7          |
| 14             | —                  | —              | 200.2              | 44.4           | 244.0              | 97.8           |
| 15             | —                  | —              | 207.6              | 45.9           | 247.9              | 93.2           |
| 16             | —                  | —              | 216.0              | 46.3           | 256.2              | 91.6           |
| 17             | —                  | —              | 227.6              | 49.7           | 259.5              | 84.0           |
| 18             | —                  | —              | 231.0              | 46.3           | 264.5              | 79.6           |
| 20             | —                  | —              | 241.0              | 43.3           | 272.9              | 71.1           |
| 22             | —                  | —              | 253.4              | 44.1           | 278.4              | 64.3           |
| 24             | —                  | —              | —                  | —              | 290.1              | 62.4           |



**Table 4. Modification factors for the loads given in tables 2 and 3**

| Species              | Modification factor                        |                                                                                               |
|----------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------|
|                      | Ultimate load<br>0.6 m from<br>top of pole | Load per millimetre<br>of deflection at point<br>of application of load<br>and crippling load |
| Redwood*             | 1.00                                       | 1.00                                                                                          |
| Corsican pine        | 1.00                                       | 1.08                                                                                          |
| Douglas fir          | 1.17                                       | 1.22                                                                                          |
| Larch:               |                                            |                                                                                               |
| Dunkeld              | 1.08                                       | 1.00                                                                                          |
| European             | 1.32                                       | 1.31                                                                                          |
| Japanese             | 1.20                                       | 1.13                                                                                          |
| Lodgepole pine       | 0.83                                       | 0.87                                                                                          |
| Norway spruce        | 0.84                                       | 0.92                                                                                          |
| Sitka spruce         | 0.82                                       | 0.85                                                                                          |
| Western red<br>cedar | 0.77                                       | 0.76                                                                                          |

\* Factors based upon  $53.8 \text{ N/mm}^2$  and  $10\,480 \text{ N/mm}^2$  given in A.2.

NOTE 1. These modification factors are based on tests carried out in accordance with appendix D and which have been carried out on 6 m lengths.

NOTE 2. The factors given are for green timber.

## Appendix B. Determination of moisture content by the oven test method

### B.1 Sampling

Samples of 75 mm or the full depth of the sapwood, whichever is the greater, shall be taken from the poles, at a point not less than 1.5 m from either end of the pole, by means of a test borer consisting of a hollow auger and extractor. The poles shall be carefully plugged, preferably with preservative impregnated beech plugs, as soon as the samples have been extracted.

### B.2 Procedure

Weigh the sample ( $m_1$ ) immediately after extraction. Dry the samples in an oven at a temperature of  $103 \pm 2^\circ\text{C}$  until the mass is constant, and again weigh immediately after removal from the drying oven ( $m_2$ ).

### B.3 Calculation

The moisture content ( $\omega$ ) expressed as a percentage of the oven dry mass is calculated from the equation:

$$\omega = \frac{m_1 - m_2}{m_2} \times 100$$

## Appendix C. Determination of the extent of preservative penetration

### C.1 Sampling

A sample shall be taken of 10 poles or 10 % of the number of poles in the charge, whichever is the greater. Fractions of a pole are to count as one pole. If there are less than 10 poles in the charge, every pole shall be bored. The poles shall either be selected and marked before treatment or be taken at random afterwards from reasonably accessible positions in the charge.

### C.2 Procedure

After treatment with preservative, take a boring or borings with a test borer not less than 1.5 m from either end of each pole of a sample of the poles in the charge.

Plug the poles as soon as the borings have been extracted. Determine the extent of penetration of the preservative either visually in the case of creosoted poles, or in accordance with BS 5666 : Part 2 in the case of CCA treated poles.

NOTE. Where necessary, a recognized chemical method of staining (see BS 5666 : Part 2) may be used to distinguish the sapwood from the heartwood.

If the borings show that the sapwood is completely penetrated in all the sample poles, all poles in the charge are deemed to comply with this clause.

If the sapwood of one or more poles from the sample is not completely penetrated, re-treat these poles and, if the number in the charge is sufficient, bore a further sample of poles equal in number to the original sample. If all the borings from this second sample show that the sapwood is penetrated completely, the remainder of the poles in the charge are deemed to comply with this clause.

If the second sample includes one or more poles where the sapwood is not completely penetrated, re-treat these poles. Continue sampling until a sample is obtained in which the sapwood of every pole is completely penetrated, or until every pole in the charge has been bored.

Subject re-treated poles to the same sampling and testing procedure as freshly treated poles.



## Appendix D. Determination of strength characteristics of poles

### D.1 Introduction

This appendix gives a method of test for the determination of the bending strength, modulus of elasticity and density of round, tapered solid wood poles. Reference is also made to the measurement of dimensions, moisture content, relative density, rate of growth and strength reducing characteristics. The method of test is based on ASTM D1036 — 58 of the American Society for Testing and Materials and on procedures previously used for the testing of transmission poles in the United Kingdom.

### D.2 General

Poles to be tested should be debarked, trimmed, notched, conditioned, and if so specified, preservative treated to simulate their usual service condition. The moisture content of the poles to be tested should be at or above their fibre saturation point.

For the duration of the test programme, untested poles should be stored under cover and clear of the ground.

Before testing, each specimen should be assigned an identification number or mark.

### D.3 Initial measurements

Before testing commences the following data should be recorded:

- (a) mass of the pole (in kg);
- (b) length of the pole (in mm);
- (c) cross-sectional sizes (in mm).

The circumference should be measured at the following positions:

- (1) butt;
  - (2) butt support;
  - (3) point of application of the test load (i.e. groundline, see figure 2);
  - (4) 1.5 m from the butt (this may coincide with the groundline (see item (3)));
  - (5) midway between the point of application of the test load and the tip support;
  - (6) tip support;
  - (7) tip.
- (d) potential strength reducing features, such as fissures, spiral grain, knots, mechanical damage and crook.

### D.4 Test conditions

The pole should be supported horizontally on rocker supports over a span of specified actual length less 600 mm with an overhang of 300 mm at the butt support (see figure 2).

The pole should be supported in its natural fall position\* to minimize any tendency for the pole to rotate when the load is applied.

To minimize indentation a steel plate 150 mm x 150 mm x 12 mm should be inserted between the specimen and the butt support and rebated into the specimen.

### D.5 Procedure

The test should be carried out as follows.

- (a) Apply load at the groundline and at a continuous rate such that failure of the pole is achieved within  $900 \pm 300$  s.
- (b) Record the deflection of the neutral axis at the loading position and measure relative to the neutral axis at the butt support.
- (c) Record a continuous automatic load/deflection curve for at least 25 % of the anticipated load, measuring loads and deflections to an accuracy of not less than 2 %.

### D.6 Calculations

#### D.6.1 Maximum stress at the groundline

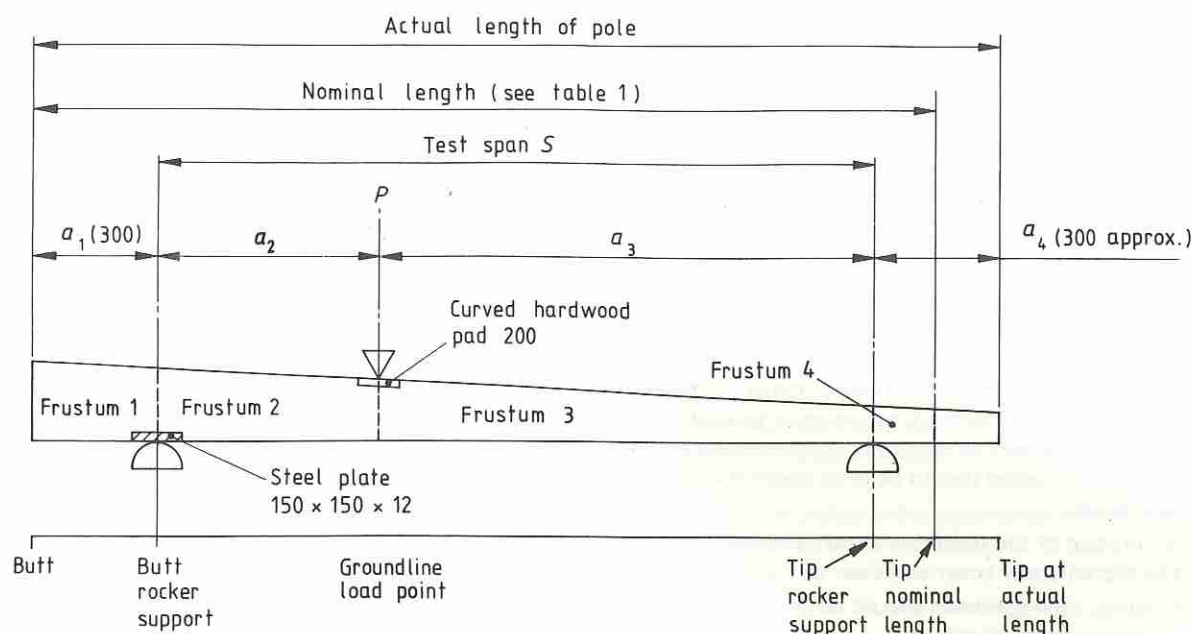
The maximum stress at the groundline  $f_g$  (in  $\text{N/mm}^2$ ) should be calculated from the following expression:

$$f_g = \frac{32}{\pi d^3} \left\{ R_t a_3 - W_3 g_3 - W_4 (a_3 + g_4) + P a_2 a_3 / S \right\}$$

where

- $d$  is the pole diameter at load point (in mm);
- $a_2$  is the distance from butt support to load point (in mm);
- $a_3$  is the distance from load point to tip support (in mm);
- $S$  is the distance between support points (in mm);
- $g_3$  is the distance from load point to centre of gravity of section between load point and tip support (in mm);
- $g_4$  is the distance from tip support to centre of gravity of overhang at tip end (in mm);

\* This should be located before test by gently rolling the pole on trestles at the butt and tip support positions.



All sizes are in millimetres.

NOTE. Not to scale.

Figure 2. Testing position of a pole (see D.6)

$W_3$  is the downward force (weight) of section between load point and tip support (in N);

$W_4$  is the downward force (weight) of overhang at tip end (in N);

$P$  is the maximum imposed load (in N);

$R_t$  is the reaction at tip support due to pole weight (in N) given by the following expression:

$$R_t = \frac{1}{S} \left\{ W_2 g_2 + (a_2 + g_3) W_3 + W_4 (a_2 + a_3 + g_4) - W_1 (a_1 - g_1) \right\}$$

where

$W_1$  is the downward force (weight) of overhang at butt end (in N);

$W_2$  is the downward force (weight) of section between butt support and load point (in N);

$g_1$  is the distance from butt to centre of gravity of overhang at butt end (in mm);

$g_2$  is the distance from butt support to centre of gravity of section between butt support and load point (in mm);

$a_1$  is the distance from butt to butt support (in mm).

The centre of gravity of a frustum of a cone  $g$  (in mm) from the base of the frustum is given by the expression:

$$g = \frac{h}{4} \frac{(d_b^2 + 3d_t^2 + 2d_b d_t)}{(d_b^2 + d_t^2 + d_b d_t)}$$

where

$d_b$  is the base diameter (in mm);

$d_t$  is the top diameter (in mm);

$h$  is the height of the appropriate frustum (in mm).

#### D.6.2 Modulus of elasticity

The modulus of elasticity  $E$  (in N/mm<sup>2</sup>) should be calculated from the following expression:

$$E = \frac{64a_3^3 a_2}{3\pi} \left( \frac{a_2}{d^2 d_s S (dS - d_s a_2)} \right) \frac{\Delta P}{\Delta \delta}$$

where

$a_2, a_3, d$  and  $S$  are as defined in D.6.1.;

$d_s$  is the pole diameter at tip support (in mm);

$\Delta P/\Delta \delta$  is the slope of load/deflection curve (in N/mm).



### D.6.3 Density

The pole should be considered as four longitudinally adjacent frustums of cones joined at the butt support, the groundline and the tip support.

The density  $\rho$  (in  $\text{kg/m}^3$ ) should be calculated from the following expression:

$$\rho = \frac{12m}{\pi \sum_{i=1}^4 h_i (d_{t,i}^2 + d_{b,i}^2 + d_{t,i}d_{b,i})}$$

where

$d_b$ ,  $d_t$  and  $h$  are as defined in D.6.1.;

$m$  is the mass of the pole (in kg).

## D.7 Other tests

### D.7.1 General

When determining the strength characteristics of the pole, the tests given in D.7.2 to D.7.4 should also be taken into account.

### D.7.2 Moisture content

Immediately after completion of the strength test an approximately 30 mm thick cross-sectional disc should be cut from the vicinity of the fracture. The disc should be free from knots, resin pockets and other abnormal features.

The moisture content should be determined in accordance with ISO 3130 for a specimen cut from the disc.

### D.7.3 Relative density

The specimen extracted for determination of moisture content (or a similar specimen cut from the same disc) should be used for the determination of relative density in accordance with ISO 3131. The calculation of relative density is based on oven-dry mass and wet volume of the specimen.

### D.7.4 Rate of growth

The average rate(s) of growth should be measured on the disc in accordance with D.7.2 and D.7.3.

The average rate of growth should be taken as the number of rings per 25 mm between radii 75 mm and 100 mm from the pith.

The total number of rings (i.e. the approximate age of the tree) should also be recorded for round poles.

## D.8 Test report

### D.8.1 General

The test report should contain details of the test material, the procedure and the results (see D.8.2 to D.8.4).

NOTE. When a programme of tests has been carried out the report may also include the maximum, minimum, mean and standard deviation values calculated for the maximum groundline stress, modulus of elasticity and relative density. However, it is not a purpose of this standard to recommend an interpretation of those statistics.

### D.8.2 Test material

The following information should be recorded:

- (a) species;
- (b) length (in m);
- (c) mass (in kg);
- (d) butt circumference (in mm);
- (e) circumference at butt support (in mm);
- (f) circumference 1.5 m from butt (in mm);
- (g) circumference at the groundline (if different from (f)) (in mm);
- (h) circumference midway between groundline and tip support (in mm);
- (i) circumference at tip support (in mm);
- (j) tip circumference (in mm).

### D.8.3 Procedure

The following information should be recorded:

- (a) the equipment used;
- (b) the temperature (in  $^{\circ}\text{C}$ ) and the relative humidity at the time of test;
- (c) any other information that may have influenced the test results.

### D.8.4 Results

The following information should be recorded:

- (a) maximum load applied (in N);
- (b) maximum groundline stress (in  $\text{N/mm}^2$ );
- (c) modulus of elasticity (in  $\text{N/mm}^2$ );
- (d) density (in  $\text{kg/m}^3$ );
- (e) the percentage moisture content;
- (f) relative density (in  $\text{kg/m}^3$ );
- (g) rate(s) of growth (in rings per 25 mm);
- (h) age (in years);
- (i) mode of failure;
- (j) any other relevant information that may influence the use of the results; for example, the growth characteristics referred to in D.3.

1. The first part of the report is a general introduction to the subject of the study. It discusses the importance of the study and the objectives of the research. It also provides a brief overview of the methodology used in the study.

2. The second part of the report is a detailed description of the study area. It includes information about the location of the study area, the population of the study area, and the characteristics of the study area. It also discusses the data sources used in the study.

3. The third part of the report is a detailed description of the study results. It includes information about the findings of the study, the conclusions drawn from the findings, and the implications of the findings. It also discusses the limitations of the study and the need for further research.

4. The fourth part of the report is a conclusion and recommendations section. It summarizes the main findings of the study and provides recommendations for future research and policy. It also discusses the significance of the study and the contribution it has made to the field.



### **Publications referred to**

- BS 144 Coal tar creosote for the preservation of timber
- BS 565 Glossary of terms relating to timber and woodwork
- BS 913 Wood preservation by means of pressure creosoting
- BS 4072 Wood preservation by means of water-borne copper/chrome/arsenic compositions
- BS 4261 Glossary of terms relating to timber preservation
- BS 5374 Specification for codes for the representation of names of countries
- BS 5666 Methods of analysis of wood preservatives and treated timber  
Part 2 Qualitative analysis
- BS 6100 Glossary of building and civil engineering terms  
Section 4.1 Characteristics and properties of timber and wood based panel products
- ISO 3130 Wood — Determination of moisture content for physical and mechanical tests
- ISO 3131 Wood — Determination of density for physical and mechanical tests
- ASTM D1036 58 Static tests of wood poles

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